SECTION A
[40 marks]

Answer **all** *the questions in this section.* **All** *questions carry* **equal** *marks.*

1. An intercity shuttle travels between two cities, Amanfro and Kotoku, covering the **first** half of the distanc at an average speed of 60 km/h and the **second** half at 90 km/h. The total distance between the two cities is 300 km.

   (*a*) Calculate, for the entire journey, the average speed of the shuttle.

   (*b*) If the speed of the shuttle in the **first** half of the journey is reduced to 50 km/h, determine its effe on the overall time.

2. A stock broker, Kojo, invested some amount in two accounts which pay 11 % and 10 % simple interest annually.The amount invested at 11 % is 140 % of the amount invested at 10 %. After $3\frac{1}{2}$ years, he earned a total of GH¢ 22,580.60 as interest. How much was invested at 11 %?

3. A contractor builds a rainwater collection system in a shape of a hemispherical tank of radius 2 m to irrigate a 500 m^2 farm. Each square metre of the farm requires 50 litres of water. Determine, with a mathematical statement, if a full tank of water would be sufficient to irrigate the farm? [Take $\pi = \frac{22}{7}$]

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4. From the top of a lighthouse 100 m high, the angles of depression of two fishing vessels are observed at 48° and 36°. The foot of the lighthouse and the sea level are on the same horizontal and the two vessels are on the same side of the lighthouse.

   (*a*) Illustrate the information in a diagram.

   (*b*) Find, correct to the **nearest** metre, the distance between the two vessels.

5. The midpoint of the line joining Y(2*a*, 3) and Z(−4, *b*) is M(1, 2*a* − 1). Find:

   (*a*) the values of *a* and *b*;

   (*b*) |YZ|.

[60 marks]

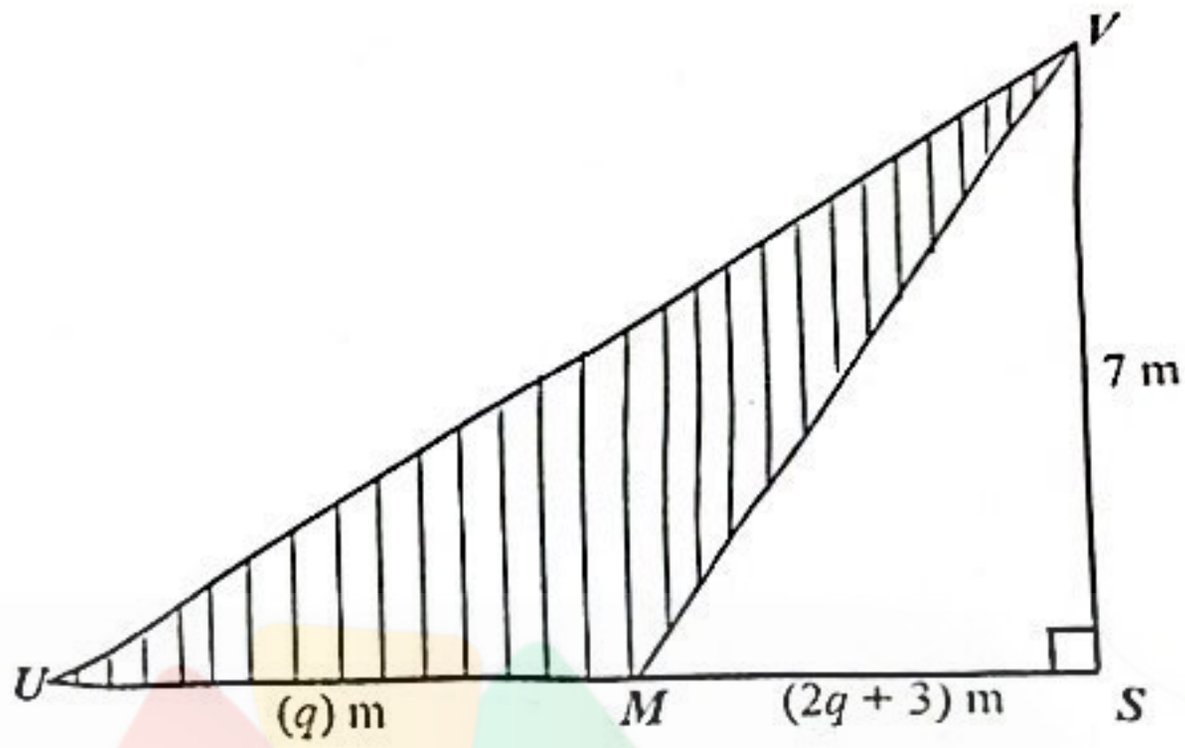
Answer **five** *questions* **only** *from this section.* **All** *questions carry* **equal** *marks.*

6. The estimated cost of building a restaurant was \$64,000.00. It was made up of the cost of labour, materials and the contractor's charges in the ratio 12 : 15 : 5 respectively. At the start of the construction, the cost of labour increased by x %, materials increased by $2x$ % due to inflation but the contractor's charges remained same. After the increase, the cost of labour was **two-thirds** the cost of materials. Determine the:

(*a*) value of x;

(*b*) new cost of building the restaurant.

7.



NOT DRAWN TO SCALE

The shaded region of the figure is a pattern to be produced in a steel casting shop. If the area of ΔSUV is 98 m^2, calculate, correct to one decimal place, the:

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(*a*) value of q;

(*b*) area of the pattern;

(*c*) perimeter of the pattern.

8. (*a*) Mr Afenyo planned to plough 120 acres of farmland each day. Due to unforeseen circumstanc 85 acres could be ploughed each day, resulting in an increase in the planned time by 2 days with 40 acres of the land still remained to be ploughed. Calculate the:

(i) number of days Mr Afenyo planned to plough the land initially;
(ii) total area of the farmland.

(*b*) A town, Mempeasem, is 5° N and another town, Mangoase is at 6.8° N on the earth's surfa Find, correct to the **nearest** whole number, the distance between the two towns.

[Take $\pi = \frac{22}{7}$, $R = 6{,}400$ km]

9. A telecommunication mast and a pillar stand on the same horizontal level. From the midpoint, O, between the foot of the pillar, S, and the foot of the mast, T, the angle of elevation of the top of the mast G, is $70°$. From the point, O, the angle of elevation of the top of the pillar, H, is $28°$. If the pillar is 10 m high:

(*a*) represent the information in a diagram;

(*b*) calculate, correct to the **nearest** whole number:
(i) the height of the mast;
(ii) $|HG|$.

10. (*a*) The points M, N and L lie on a circle with centre O. An angle subtended at the centre by the minor arc MN is $88°$.
(i) Illustrate the information in a diagram.
(ii) If L is a point on the minor arc, determine the value of the angle subtended by the chord MN at L.

(*b*) A tiler, Coffie, is able to lay 400 tiles each day in 8 hours. Another tiler, Boakye, works at $\frac{3}{2}$ times Coffie's rate when laying same kind of tiles.
(i) How many tiles can Boakye lay in 8 hours?
(ii) If both Coffie and Boakye decide to work together, determine the time required to lay 2000 tiles.

11. A cardboard is in the form of an isosceles triangle with vertices ABC. $|AB| = |AC| = (2p + q)$ m, $|BC| = (3p + 5)$ m and the perimeter of the cardboard is $(7q + 3)$ m.

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(*a*) Illustrate the information in a diagram.

(*b*) Given that $p : q = 2 : 3$, find, correct to one decimal place, the:
(i) value of p and q;
(ii) area of the cardboard.

12. The following are the marks obtained by 50 students in an English test.

27	31	35	52	81	50	48	15	59	6
17	41	30	94	23	38	40	70	76	18
52	61	43	68	56	57	62	82	54	25
18	35	9	74	59	46	71	4	74	33
61	16	50	45	41	45	28	22	26	45

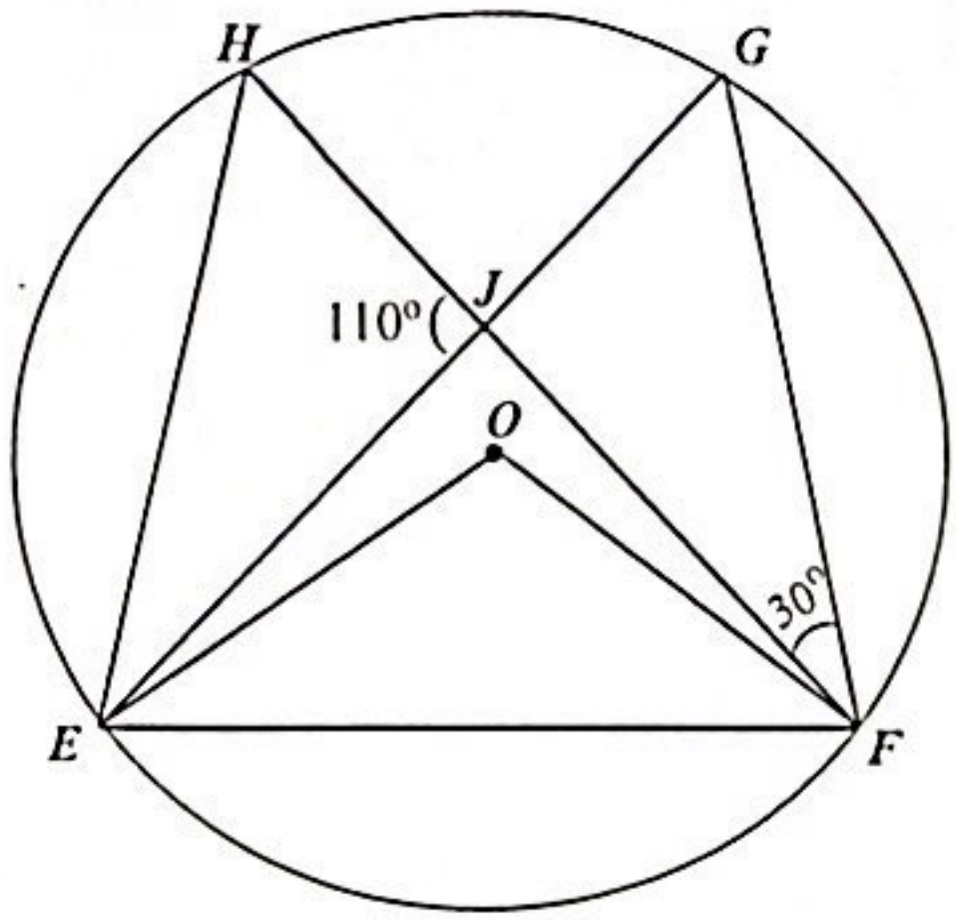
(*a*) Using the class interval, 1 – 10, 11 – 20, 21 – 30, ... , construct a cumulative frequency table for the distribution.

(*b*) Draw a cumulative frequency curve for the distribution.

(*c*) Use the curve to find the upper quartile.

(*d*) If only 10 % of the students passed the test, use the curve to determine the pass mark.

13. (*a*)



NOT DRAWN TO SCALE

In the diagram, E, F, G and H lie on the circumference of the circle centre O. $\overline{EG}$ and $\overline{FH}$ intersect at J such that $\angle GFH = 30°$ and $\angle HJE = 110°$. Calculate:

(i) $\angle EOF$;

(ii) $\angle JEH$.

(*b*) Consider the following statements.

u: Most Arts students are good at debates.

v: Some Non-Arts students are also good at debates.

(i) Illustrate the information in a Venn diagram.

(ii) Determine from the diagram whether the following conclusions are **Valid** or **Not Valid**.

(α) John is a non-Arts student so he is **not** good at debates.

(β) Aku is an Arts student so she is good at debates.

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